

Boundary-Localized Genomic Utility and Clinical-Tail Dominance: A Likelihood-Ratio Theory for ZeBRA and Fibrotic Interstitial Lung Disease

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Abstract—We consider two information channels for predicting future fibrotic interstitial lung disease (ILD): a static germline genomic channel and a longitudinal clinical-history channel summarized by the Zero-burden Risk Assessment (ZeBRA). Preliminary analyses in an adjudicated fibrotic ILD/fibrotic interstitial lung abnormality cohort show a characteristic pattern: ZeBRA has strong global discrimination, adding many genomic features produces little global AUC gain, yet the high-effect MUC5B promoter variant rs35705950 can improve sensitivity in restricted ZeBRA score regions at essentially matched false-positive rate. We show that this pattern follows from a general likelihood-ratio argument. For a non-Mendelian disease, genomic evidence is bounded, whereas a longitudinal clinical sequence can accumulate likelihood evidence. Bayes factorization implies that genomic information can change a likelihood-ratio decision only in a bounded neighborhood of the clinical decision boundary; the width of this neighborhood is fixed on the log-likelihood scale by the strength of genomic evidence. By the Neyman–Pearson lemma, a clinical likelihood-ratio test is optimal among clinical-history tests at fixed false-positive rate. If the clinical likelihood ratio has an extended upper tail while genomic likelihood ratios are bounded, then at sufficiently extreme specificity the clinical channel strictly dominates every genomic-only rule. These results provide a formal explanation for why genomics may add little to global discrimination while remaining useful in selected boundary-adjacent patients.

Index Terms—ZeBRA, fibrotic interstitial lung disease, idiopathic pulmonary fibrosis, MUC5B, genomics, likelihood ratio, Neyman–Pearson lemma, Bayesian inference, decision boundary.

I. Motivating problem: ZeBRA and genomic risk in fibrotic ILD

Fibrotic ILDs, including idiopathic pulmonary fibrosis (IPF), have a strong but non-Mendelian genetic component. The MUC5B promoter variant rs35705950 is among the strongest common genetic risk factors for pulmonary fibrosis [2]–[4]. In the original report, the minor allele was present much more frequently among subjects with IPF than among controls [2]. At the same time, disease emergence is accompanied by a growing sequence of clinically observable diagnoses, prescriptions, procedures, and comorbidities. ZeBRA is designed to summarize such routinely observed longitudinal clinical histories into a future-risk score without requiring laboratory tests, imaging, or questionnaires [1].

Our recent fibrotic ILD analyses ask whether genomic information improves prediction once ZeBRA is known. The empirical picture is not the one expected from a simple additive model. In repeated-split analyses of the adjudicated endpoint “FILD or FILA ADJUDICATED,” ZeBRA alone has AUC approximately 0.81–0.82. Adding the available genomic variables with regularized discriminative models changes global AUC by only about 10^{-3} . ZeBRA itself does not reconstruct MUC5B carrier status: the observed AUC for predicting rs35705950 carrier status from ZeBRA is essentially 0.5. Yet MUC5B is enriched among some ZeBRA false negatives, and a rule that uses MUC5B within selected ZeBRA-score regions improves sensitivity at a clinically relevant operating point. In one repeated-split analysis near a nominal 5% false-positive-rate (FPR) operating point, sensitivity increased from approximately 45.8% to 49.7% at essentially unchanged realized FPR, with LR+ increasing from about 7.9 to 8.5 and LR– decreasing from about 0.58 to 0.54. Local MUC5B effects were particularly strong in selected intermediate-high ZeBRA percentile bands. These observations are exploratory and the particular percentile bands were selected post hoc; they motivate the theory below rather than constitute independent confirmation of it.

The apparent paradox is therefore:

A strong genomic risk factor can be biologically important, and can improve decisions for selected individuals, while contributing almost nothing to global discrimination once a strong clinical-history predictor is available.

We show that this behavior is expected under a general Bayesian likelihood-ratio formulation.

The archived three-target AUC comparison and the repeated-split boundary analyses are reported explicitly in Section VII and Supplementary Tables III–VI.

II. Latent-state and information-channel formulation

Let B_t denote the latent biological state at time t , G the germline genomic state, and $E_{0:t}$ the accumulated non-genetic exposure history. The biological state evolves under

the influence of G and $E_{0:t}$. For a fixed future horizon $h > 0$, define the disease-risk functional

$$\rho_h(B_t) = \mathbb{P}(T \leq t + h \mid T > t, B_t), \quad (1)$$

where T is the disease-event time. The clinical event sequence $C_{0:t}$ is a noisy observation of the latent biological trajectory $B_{0:t}$. In the idealized Bayesian limit, a clinical-history predictor satisfies

$$Z_h(C_{0:t}) = \mathbb{P}(T \leq t+h \mid C_{0:t}) = \int \rho_h(b) d\mathbb{P}(B_t = b \mid C_{0:t}). \quad (2)$$

Thus ZeBRA need not reconstruct B_t ; it estimates the disease-relevant functional of B_t from the clinical observation channel.

For the theorem it is enough to reduce the problem to binary prediction. Let

$$Y = \mathbf{1}\{T \leq t + h\} \in \{0, 1\}, \quad (3)$$

and write C for the available clinical history. The genomic channel G and clinical channel C are then placed on equal statistical footing through their disease-versus-control likelihood ratios.

III. Likelihood-ratio representation

Let $\mathbb{P}_1(\cdot) = \mathbb{P}(\cdot \mid Y = 1)$ and $\mathbb{P}_0(\cdot) = \mathbb{P}(\cdot \mid Y = 0)$. Assume the relevant distributions admit densities with respect to common dominating measures. Define the clinical likelihood ratio

$$\Lambda_C(c) = \frac{p(c \mid Y = 1)}{p(c \mid Y = 0)}, \quad (4)$$

and the residual genomic likelihood ratio conditional on clinical history

$$\Lambda_{G|C}(g; c) = \frac{p(g \mid Y = 1, C = c)}{p(g \mid Y = 0, C = c)}. \quad (5)$$

The full joint likelihood ratio is

$$\Lambda_{C,G}(c, g) = \frac{p(c, g \mid Y = 1)}{p(c, g \mid Y = 0)}. \quad (6)$$

By the probability chain rule,

$$\boxed{\Lambda_{C,G}(c, g) = \Lambda_C(c) \Lambda_{G|C}(g; c).} \quad (7)$$

No conditional-independence assumption is required.

If Z is a calibrated clinical posterior, $Z = \mathbb{P}(Y = 1 \mid C)$, and $\pi = \mathbb{P}(Y = 1)$, Bayes' rule yields

$$\frac{Z}{1 - Z} = \frac{\pi}{1 - \pi} \Lambda_C(C), \quad (8)$$

so thresholding calibrated ZeBRA is equivalent to thresholding Λ_C . More generally, the same results apply whenever the clinical score is a strictly monotone transformation of Λ_C .

IV. Main theorem

The key biological assumptions are deliberately weak.

A1 (Non-Mendelian genomic evidence). No measured germline genomic state determines disease with probability one or zero. Equivalently, genomic Bayes factors are finite. We assume constants $0 < m \leq 1 \leq M < \infty$ such that

$$0 < m \leq \Lambda_{G|C}(g; c) \leq M < \infty \quad (9)$$

for relevant (g, c) , and that the marginal genomic likelihood ratio

$$\Lambda_G(g) = \frac{p(g \mid Y = 1)}{p(g \mid Y = 0)} \quad (10)$$

is bounded above by $M_G < \infty$.

A2 (Accumulating clinical evidence). Clinical history is sequential, $C_n = (X_1, \dots, X_n)$, and continued diagnoses, prescriptions, procedures, and comorbidities can produce an extended disease-evidence tail. Formally, Λ_C is not bounded above on the relevant sequence of clinical histories: for arbitrarily large K there are events with positive $\mathbb{P}_0$ probability on which $\Lambda_C \geq K$.

Theorem 1 (Boundary-localized genomic utility and clinical-tail dominance). Under A1–A2:

(i) Exact factorization. The optimal joint disease-versus-control likelihood ratio is

$$\Lambda_{C,G} = \Lambda_C \Lambda_{G|C}. \quad (11)$$

(ii) Boundary localization. For any likelihood-ratio decision threshold $\lambda > 0$, genomic information can change which side of the threshold an observation occupies only if

$$\boxed{\frac{\lambda}{M} \leq \Lambda_C(c) \leq \frac{\lambda}{m}.} \quad (12)$$

Equivalently,

$$\boxed{-\log M \leq \log \Lambda_C(c) - \log \lambda \leq -\log m.} \quad (13)$$

Hence the genomic decision-relevant region has fixed width

$$\log(M/m) \quad (14)$$

on the clinical log-likelihood scale, independent of where the decision boundary λ is placed.

(iii) Genomic-only power is bounded. For any test $\phi(G) \in [0, 1]$ using genomics alone, with false-positive rate $\alpha = \mathbb{E}_0[\phi(G)]$, its true-positive rate satisfies

$$\boxed{\mathbb{E}_1[\phi(G)] \leq M_G \alpha.} \quad (15)$$

Thus the positive likelihood ratio of any genomic-only decision rule is at most M_G .

(iv) Clinical superiority at extreme specificity. For any $K > M_G$ with $\alpha_K = \mathbb{P}_0(\Lambda_C \geq K) > 0$, the clinical likelihood-ratio test

$$\phi_K(C) = \mathbf{1}\{\Lambda_C(C) \geq K\} \quad (16)$$

has

$$\frac{\mathbb{P}_1(\Lambda_C \geq K)}{\mathbb{P}_0(\Lambda_C \geq K)} \geq K > M_G. \quad (17)$$

Therefore, at the same FPR α_K , its power is strictly greater than that of every genomic-only test. Moreover,

$$\alpha_K \leq \frac{1}{K} \rightarrow 0 \quad (18)$$

as $K \rightarrow \infty$. Hence the superiority occurs in the extreme-specificity limit.

(v) Clinical-tail dominance in the combined evidence. Along sequences with $\Lambda_C \rightarrow \infty$,

$$\log \Lambda_{C,G} = \log \Lambda_C + O(1), \quad (19)$$

and therefore

$$\boxed{\frac{\log \Lambda_{C,G}}{\log \Lambda_C} \rightarrow 1.} \quad (20)$$

Thus bounded genomic evidence is asymptotically negligible relative to the accumulated clinical log-likelihood in the extreme clinical-evidence tail.

Proof. Part (i) is the chain rule:

$$\Lambda_{C,G}(c, g) = \frac{p(c, g | Y = 1)}{p(c, g | Y = 0)} \quad (21)$$

$$= \frac{p(c | Y = 1)p(g | Y = 1, c)}{p(c | Y = 0)p(g | Y = 0, c)} \quad (22)$$

$$= \Lambda_C(c)\Lambda_{G|C}(g; c). \quad (23)$$

For part (ii), compare the clinical-only decision $\Lambda_C \geq \lambda$ with the joint decision $\Lambda_C \Lambda_{G|C} \geq \lambda$. If

$$\Lambda_C < \lambda/M, \quad (24)$$

then A1 gives

$$\Lambda_C \Lambda_{G|C} \leq \Lambda_C M < \lambda, \quad (25)$$

so no genomic state can move the observation above the threshold. Conversely, if

$$\Lambda_C > \lambda/m, \quad (26)$$

then

$$\Lambda_C \Lambda_{G|C} \geq \Lambda_C m > \lambda, \quad (27)$$

so no genomic state can move the observation below the threshold. Hence a threshold change is possible only on the interval in (12). Taking logarithms gives (13); its width is

$$(-\log m) - (-\log M) = \log(M/m), \quad (28)$$

which does not depend on λ .

For part (iii), let $\phi(G) \in [0, 1]$. Using the Radon–Nikodym identity under $\mathbb{P}_0$,

$$\mathbb{E}_1[\phi(G)] = \mathbb{E}_0[\Lambda_G(G)\phi(G)] \quad (29)$$

$$\leq M_G \mathbb{E}_0[\phi(G)] \quad (30)$$

$$= M_G \alpha. \quad (31)$$

This proves (15). The result holds for randomized as well as deterministic genomic rules.

For part (iv), define $A_K = \{c : \Lambda_C(c) \geq K\}$. Then

$$\mathbb{P}_1(A_K) = \mathbb{E}_0[\Lambda_C(C)\mathbf{1}\{A_K\}] \quad (32)$$

$$\geq K \mathbb{E}_0[\mathbf{1}\{A_K\}] \quad (33)$$

$$= K \mathbb{P}_0(A_K) \quad (34)$$

$$= K \alpha_K. \quad (35)$$

Therefore the clinical rule has LR+ at least K . If $K > M_G$, any genomic-only test with the same FPR α_K has power at most $M_G \alpha_K < K \alpha_K \leq \mathbb{P}_1(A_K)$. By the Neyman–Pearson lemma [5], thresholding Λ_C is also the most powerful rule among all tests measurable with respect to C at the corresponding size (with the usual boundary randomization if needed).

Finally, because $\mathbb{E}_0[\Lambda_C] = 1$, Markov’s inequality gives

$$\alpha_K = \mathbb{P}_0(\Lambda_C \geq K) \leq \frac{\mathbb{E}_0[\Lambda_C]}{K} = \frac{1}{K}, \quad (36)$$

so $\alpha_K \rightarrow 0$ as $K \rightarrow \infty$. This is precisely the high-specificity regime.

For part (v), A1 implies

$$\log m \leq \log \Lambda_{G|C} \leq \log M, \quad (37)$$

so $\log \Lambda_{G|C} = O(1)$. By part (i),

$$\log \Lambda_{C,G} = \log \Lambda_C + \log \Lambda_{G|C} = \log \Lambda_C + O(1). \quad (38)$$

Dividing by $\log \Lambda_C$ and letting $\Lambda_C \rightarrow \infty$ yields (20). $\square$

Corollary 1 (Posterior-risk form for ZeBRA). If ZeBRA is calibrated so that $Z = \mathbb{P}(Y = 1 | C)$, let $\tau \in (0, 1)$ be any ZeBRA decision threshold and define posterior-odds threshold $\kappa = \tau/(1 - \tau)$. Then genomic information can change the decision only if

$$\boxed{\frac{\kappa}{M} \leq \frac{Z}{1 - Z} \leq \frac{\kappa}{m}.} \quad (39)$$

Equivalently,

$$-\log M \leq \text{logit}(Z) - \text{logit}(\tau) \leq -\log m. \quad (40)$$

Thus genomic utility is localized around whatever ZeBRA operating boundary is chosen.

Proof. Equation (8) shows that ZeBRA posterior odds differ from Λ_C by the positive constant prior odds $\pi/(1 - \pi)$. Substitution into Theorem 1(ii) yields the result. $\square$

Corollary 2 (Any genomic model inherits the bound). Let $S = f(G)$ be any measurable genomic predictor, including a polygenic risk score, gradient-boosted model, random forest, neural network, or generative genomic model. If the full genomic likelihood ratio is bounded, then the likelihood ratio of S is also bounded. Therefore no deterministic compression or machine-learning transformation of the same genomic information can evade the finite-evidence and extreme-specificity conclusions of Theorem 1.

Proof. For a score value s , the score likelihood ratio can be written as a conditional average of the full genomic likelihood ratio under $\mathbb{P}_0$:

$$\Lambda_S(s) = \mathbb{E}_0[\Lambda_G(G) | S = s]. \quad (41)$$

A conditional expectation of a random variable bounded above by M_G is itself bounded above by M_G . The analogous lower bound follows if Λ_G is bounded away from zero. $\square$

V. Why the clinical channel can have an extended tail

The theorem does not assume that every later clinical event is informative, nor that disease becomes perfectly observable. It requires only that the clinical channel can accumulate sufficiently strong evidence on some disease trajectories. For a sequential history $C_n = (X_1, \dots, X_n)$, the clinical likelihood ratio factorizes exactly as

$$\Lambda_C(C_n) = \prod_{k=1}^n \frac{p(X_k | Y = 1, X_{<k})}{p(X_k | Y = 0, X_{<k})}, \quad (42)$$

so that

$$\log \Lambda_C(C_n) = \sum_{k=1}^n \log \frac{p(X_k | Y = 1, X_{<k})}{p(X_k | Y = 0, X_{<k})}. \quad (43)$$

This is the standard likelihood-ratio accumulation underlying sequential testing [6]. The genomic state does not acquire new germline variants as the disease approaches, whereas the clinical history can acquire additional diagnoses, procedures, prescriptions, and comorbidities. The difference is therefore not that clinical and genomic predictors obey different probability laws; both enter Bayes' rule as evidence. The difference is that one channel is static and finite while the other is longitudinal and can supply repeated likelihood-ratio factors.

Equation (42) also clarifies the sense in which clinical odds may rise sharply: likelihood evidence adds in log space but multiplies in odds space. No claim of exponential growth follows from AUC alone. Rather, when a sequence contains repeated positive log-likelihood increments, the posterior odds are the exponential of their accumulated sum.

VI. Interpretation of the ILD/IPF observations

The theorem organizes the empirical findings into three statements.

A. Why global AUC need not improve

A strong clinical-history score can already rank most patients correctly. If the bounded genomic factor can change decisions only in the band (12), then only patients occupying that band can be materially reordered across the chosen clinical threshold. Global AUC averages over all case-control pairs, so a substantial local benefit can coexist with a negligible global AUC change. This is consistent with the observed mean AUC increment of order 10^{-3} when broad genomic features were added to ZeBRA.

B. Why MUC5B can matter locally despite weak ZeBRA-genotype association

The observation that ZeBRA cannot predict rs35705950 carrier status globally does not imply that MUC5B is irrelevant conditionally. The relevant quantity is not $I(Z; G)$ but the residual disease evidence supplied by G at a particular clinical state, represented here by $\Lambda_{G|C}$. A high-burden allele can therefore be nearly independent of ZeBRA globally yet shift selected boundary-adjacent patients

across a disease-decision threshold. This is consistent with the observed enrichment of MUC5B among selected ZeBRA false negatives and the improved matched-FPR sensitivity obtained by genomic rescue rules.

C. Why ZeBRA should dominate at very high specificity

For any genomic-only rule, Theorem 1(iii) bounds the achievable LR+ by M_G . In contrast, a clinical likelihood-ratio tail above threshold K has $\text{LR+} \geq K$, and its FPR is at most $1/K$. Therefore, once the clinical history reaches likelihood-ratio levels larger than the maximum genomic Bayes factor, the clinical channel must be superior to every genomic-only predictor at those extreme-specificity operating points. The empirical observation that ZeBRA achieves large LR+ at stringent specificity is therefore more directly relevant to this theory than AUC alone.

The expected qualitative geometry is consequently a transition from genomic susceptibility, through a boundary region in which both channels can matter, to clinical-tail dominance; it is not a uniform competition between two globally interchangeable predictors.

VII. Empirical correspondence in the v1 ILD analysis

The archived ZEBRA_GENDRIVERS_CLASSIFIER/v1 analysis used three adjudicated clinical targets and a targeted genomic panel assembled before model fitting. The three targets were (i) FILD or FILA ADJUDICATED, (ii) NONFIBROTIC ILD ADJUDICATED, and (iii) NON FIBROTIC ILA ADJUDICATED. Under the legacy notebook-03/032 construction, each target was converted to a binary endpoint and the analysis was restricted to observations with an available ZeBRA score. The resulting target-specific cohort sizes and case counts are documented in Supplementary Table III.

The genomic matrix was assembled by matching an expanded ILD/IPF locus list to available genotype columns and taking the union with the pre-existing biological-driver panel. The archived v1 matrix contains 171 distinct variant loci, represented during discriminative modeling by three categorical one-hot states per locus. Of these, 148 variant columns were matched from the expanded MORE_LOCI.csv screen before union and de-duplication with the original driver set. The complete 171-locus panel is given in Supplementary Table IV; rs35705950 (MUC5B) is included explicitly.

Table I gives the archived fixed-split AUC comparison. Across the three phenotype contexts, ZeBRA remained the stronger single channel, while adding the genomic panel did not yield a consistent global AUC improvement. The genomic-only AUCs were materially lower than the corresponding ZeBRA AUCs for all three targets.

The boundary-localization prediction is more directly tested by repeated outer splits for the FILD/FILA target. Figure 1 shows that a fixed regional MUC5B rule does not improve global AUC: over 100 splits, mean AUC was 0.8160 for raw ZeBRA and 0.8135 for the regional hybrid.

TABLE I
Archived fixed-split AUC comparison across the three v1 clinical targets.

Target	ZeBRA	Genomic	Combined
FILD/FILA	0.804	0.643	0.800
Nonfibrotic ILD	0.838	0.629	0.855
Nonfibrotic ILA	0.809	0.573	0.806

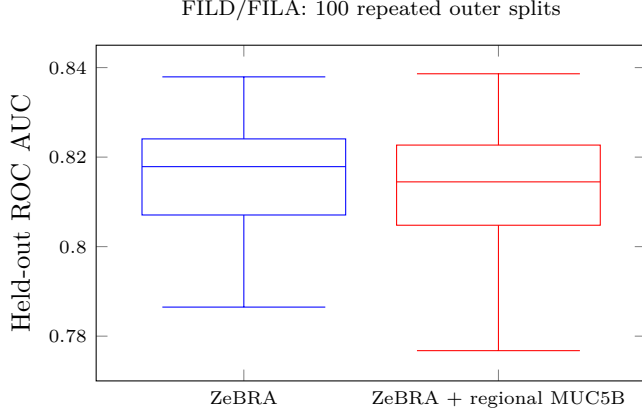


Fig. 1. Held-out AUC distributions over 100 repeated outer splits for the FILD/FILA target. The regional MUC5B rule does not improve global AUC, consistent with genomic information being useful only in a restricted decision region rather than throughout the score space.

TABLE II
Repeated-split performance near the nominal 5% FPR operating point ($n = 100$ splits).

Metric	ZeBRA	Hybrid	Difference
AUC	0.8160	0.8135	-0.0025
Sensitivity	0.4576	0.4969	+0.0393
Realized FPR	0.0586	0.0591	+0.0005
LR+	7.89	8.46	+0.57
LR-	0.576	0.535	-0.042
PPV	0.1367	0.1453	+0.0086

Nevertheless, at the nominal 5% FPR operating point the same rule increased mean held-out sensitivity from 0.4576 to 0.4969 while the realized FPR changed only from 0.0586 to 0.0591. Mean LR+ increased from 7.89 to 8.46 and mean LR- decreased from 0.576 to 0.535 (Table II). Thus the observed effect is concentrated at an operating boundary despite negligible—indeed slightly unfavorable—global AUC change.

Figure 2 shows the corresponding matched-FPR behavior across the evaluated operating points. The largest sensitivity gain appears around the intermediate 5% FPR boundary rather than in the most extreme clinical tail. This is the qualitative pattern predicted by Theorem 1: bounded genomic evidence can be useful where the clinical likelihood ratio is sufficiently close to the chosen threshold, while it has progressively less ability to alter decisions when clinical evidence is far from that threshold.

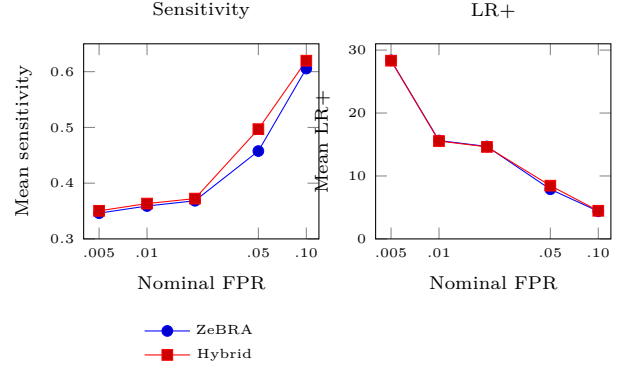


Fig. 2. Repeated-split boundary performance of raw ZeBRA and the fixed regional MUC5B hybrid. Left: mean held-out sensitivity at training-derived operating points. Right: mean LR+. The regional rule was selected exploratorily and requires independent or prespecified confirmation.

VIII. Scope and limitations

The theory is a statement about information channels, not about a specific machine-learning algorithm. It applies to the full measured genomic state and to any predictor derived from it, provided disease is non-Mendelian in the finite-likelihood-ratio sense. It also does not require independence between genomics and clinical history; dependence is explicitly retained through $\Lambda_{G|C}$.

Three qualifications are important. First, the theorem does not state that genomics changes no posterior probabilities away from the boundary; it states that bounded genomic evidence cannot change the decision away from the boundary. Second, high AUC alone does not imply an extended likelihood-ratio tail. The relevant empirical evidence is high likelihood ratio at stringent specificity, together with the sequential nature of the clinical record. Third, the current percentile-specific MUC5B rescue regions were identified exploratorily and require validation in an independent cohort or a fully prespecified held-out analysis.

IX. Conclusion

For non-Mendelian disease, static genomic evidence is finite. Longitudinal clinical evidence, by contrast, can accumulate through repeated observations. Bayes' rule factorizes the joint evidence into clinical and residual genomic likelihood-ratio factors, while the Neyman-Pearson lemma identifies likelihood-ratio thresholding as the optimal fixed-FPR decision rule. These facts imply two general consequences: genomic information can alter a clinical decision only within a bounded neighborhood of the chosen clinical decision boundary, and a clinical channel with a sufficiently extended likelihood-ratio tail must eventually dominate any bounded genomic-only predictor at extreme specificity. The preliminary ZeBRA-fibrotic ILD results—negligible global AUC gain from broad genomic augmentation, local MUC5B enrichment among false negatives, and improved sensitivity from boundary-targeted

genomic rescue—are the empirical pattern predicted by this theory.

A v1 phenotype definitions and cohort construction

The three phenotype fields were read from REPHENOTYPES FOR IC.csv. In the legacy v1 notebook construction, Y/y was mapped to 1, N/n to 0, values were converted to numeric, missing values were assigned 0 for the notebook-03/032 target geometry, and observations without an available predicted_risk were excluded. Table III records the exact stored target labels and the cohort counts reproduced by the v1 sanity split.

B Genomic panel construction

The v1 data-construction notebook combined the pre-existing biological-driver columns with loci matched from MORE_LOCI.csv. The latter file contained 3,113 candidate records; 148 genotype columns were matched to the available genomic header. After union with the original biological-driver columns and de-duplication, the archived model matrix contained 171 distinct variant loci across 19,651 rows before phenotype/ZeBRA cohort restriction. Each locus was cast to the categorical states 0, 1, and 2 and represented by one-hot indicator columns for the discriminative analyses. Supplementary Table IV lists the unique variant/allele prefixes reconstructed from the archived v1 combined-model feature matrix.

C Detailed empirical comparisons

Supplementary Table V gives the archived fixed-split AUC values used in Table I. These values are retained as a descriptive record of the v1 analysis. The genomic-only rows in the archived validation output were evaluated on their stored genomic validation geometry, whereas ZeBRA and combined values correspond to the target-specific ZeBRA-available validation geometry; the comparison should therefore be interpreted as descriptive rather than as a formal paired test.

The fixed regional rule used MUC5B information within ZeBRA percentile intervals 0–50, 65–75, and 85–95, and used the calibrated ZeBRA score outside those regions. These regions were selected after inspection of the local-information curves and are therefore exploratory. Across 100 repeated outer splits, the training-split median MUC5B odds ratios were 2.88, 4.40, and 4.07 in the three respective regions (Supplementary Fig. 3).

Supplementary Table VI reports the repeated-split operating-point results. The nominal target FPR is used only to determine a threshold in each training split; the table reports the mean realized held-out FPR. The 5% operating point shows the clearest boundary effect: sensitivity increased by 0.0393 on average while realized FPR changed by 0.0005. The mean global AUC difference was -0.0025 at every operating point because the same pair of continuous raw/hybrid scores was evaluated before thresholding.

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TABLE III
Clinical prediction targets used in the archived v1 analysis.

Stored target label	Operational interpretation in the v1 analysis	N	Positives
FILD or FILA ADJUDICATED	Adjudicated fibrotic ILD or fibrotic interstitial lung abnormality endpoint.	12,825	254
NONFIBROTIC ILD ADJUDICATED	Adjudicated nonfibrotic interstitial lung disease endpoint.	12,825	27
NON FIBROTIC ILA ADJUDICATED	Adjudicated nonfibrotic interstitial lung abnormality endpoint.	12,825	74

TABLE IV

Complete 171-locus genomic panel used by the archived v1 combined model. The suffix after the rs identifier is the allele label retained in the v1 feature name; each locus was represented by categorical states 0/1/2.

No.	Locus	No.	Locus	No.	Locus
1	rs1015097_A	58	rs17616659_G	115	rs57625989_A
2	rs10422762_A	59	rs17690703_C	116	rs601878_C
3	rs10484326_T	60	rs1851741_G	117	rs616768_G
4	rs10518693_C	61	rs1923802_T	118	rs61829240_G
5	rs10748858_T	62	rs1981997_G	119	rs61867586_C
6	rs10776482_A	63	rs2034650_A	120	rs62112917_C
7	rs10783497_A	64	rs2067832_G	121	rs62354543_A
8	rs10842996_T	65	rs2071627_C	122	rs62505367_T
9	rs10869195_A	66	rs2076295_T	123	rs62530710_G
10	rs10881263_G	67	rs2109069_G	124	rs6491201_C
11	rs10886437_C	68	rs2228972_G	125	rs6499642_C
12	rs10976897_G	69	rs2353082_C	126	rs6532150_G
13	rs11083093_G	70	rs2361278_T	127	rs664034_T
14	rs11102362_G	71	rs2375892_T	128	rs66884960_C
15	rs11154034_T	72	rs2577254_T	129	rs6736148_C
16	rs11191865_G	73	rs2609255_T	130	rs68170294_G
17	rs1123573_A	74	rs2609260_T	131	rs6899532_G
18	rs11264306_A	75	rs2726049_A	132	rs6970647_T
19	rs114064385_G	76	rs2736100_A	133	rs699240_T
20	rs116575227_A	77	rs2857476_C	134	rs7041597_T
21	rs11680126_C	78	rs28703652_G	135	rs708686_C
22	rs117427783_T	79	rs302837_A	136	rs7122936_A
23	rs117750427_A	80	rs3130157_G	137	rs71389435_C
24	rs1189320_G	81	rs3131520_C	138	rs72722911_G
25	rs12275748_C	82	rs325093_T	139	rs72755986_C
26	rs12422371_C	83	rs34262641_A	140	rs72762052_C
27	rs12446021_G	84	rs34425120_G	141	rs73008477_C
28	rs12610495_A	85	rs34547036_C	142	rs73409602_C
29	rs12626788_C	86	rs34625679_T	143	rs73997403_C
30	rs12658068_G	87	rs34825924_G	144	rs74490361_T
31	rs12665063_A	88	rs35677769_G	145	rs74760554_C
32	rs12710674_C	89	rs35705950.1_G	146	rs74987898_C
33	rs12713927_G	90	rs3744761_C	147	rs75291377_G
34	rs1273431_A	91	rs3750920_C	148	rs7547997_G
35	rs1278769_G	92	rs3778337_G	149	rs756345_G
36	rs12928193_G	93	rs3829223_C	150	rs75714781_G
37	rs12975986_C	94	rs3887876_C	151	rs75764565_C
38	rs13183751_G	95	rs4077759_T	152	rs76160568_A
39	rs13299670_C	96	rs41482151_C	153	rs76475316_T
40	rs1422506_G	97	rs415430_T	154	rs76739551_C
41	rs1423029_A	98	rs45619034_G	155	rs76931164_T
42	rs144128391_T	99	rs4672388_T	156	rs77144736_C
43	rs144948432_A	100	rs4680134_C	157	rs7725218_G
44	rs145802912_A	101	rs469942_C	158	rs7730878_A
45	rs146764446_A	102	rs4727443_C	159	rs77998999_A
46	rs1479945_C	103	rs4811028_G	160	rs78288787_C
47	rs148106529_G	104	rs4918072_G	161	rs79242376_G
48	rs1488496_C	105	rs4960330_T	162	rs79252367_G
49	rs148877274_T	106	rs55672765_C	163	rs7934606_C
50	rs1496725_A	107	rs55682827_C	164	rs7942850_T
51	rs151036604_T	108	rs55993474_A	165	rs7952154_C
52	rs1669778_T	109	rs56111559_G	166	rs893889_G
53	rs16837903_G	110	rs56184452_G	167	rs918174_T
54	rs169889_A	111	rs56387623_T	168	rs923077_A
55	rs17007563_C	112	rs56744128_A	169	rs9397724_G
56	rs17019820_A	113	rs5743890_T	170	rs9577395_C
57	rs17372089_A	114	rs5743894_T	171	rs9931202_G

TABLE V
Archived v1 fixed-split AUC values and target-specific validation counts.

Target	ZeBRA AUC	Genomic AUC	Combined AUC	ZeBRA/combined n_+	Genomic n_+	ZeBRA/combined n_-
FILD/FILA	0.8041	0.6426	0.8000	152	206	7,543
Nonfibrotic ILD	0.8379	0.6290	0.8547	16	22	7,679
Nonfibrotic ILA	0.8092	0.5728	0.8060	44	55	7,651

TABLE VI
Repeated-split FILD/FILA boundary results for the fixed regional MUC5B rule ($n = 100$ splits).

Nominal FPR	ZeBRA sens.	Hybrid sens.	Δ sens.	ZeBRA FPR	Hybrid FPR	ZeBRA LR+	Hybrid LR+
0.005	0.3462	0.3504	+0.0042	0.0202	0.0204	28.36	28.32
0.010	0.3590	0.3634	+0.0044	0.0239	0.0241	15.63	15.55
0.020	0.3684	0.3723	+0.0039	0.0263	0.0266	14.69	14.62
0.050	0.4576	0.4969	+0.0393	0.0586	0.0591	7.89	8.46
0.100	0.6056	0.6197	+0.0141	0.1427	0.1434	4.39	4.47

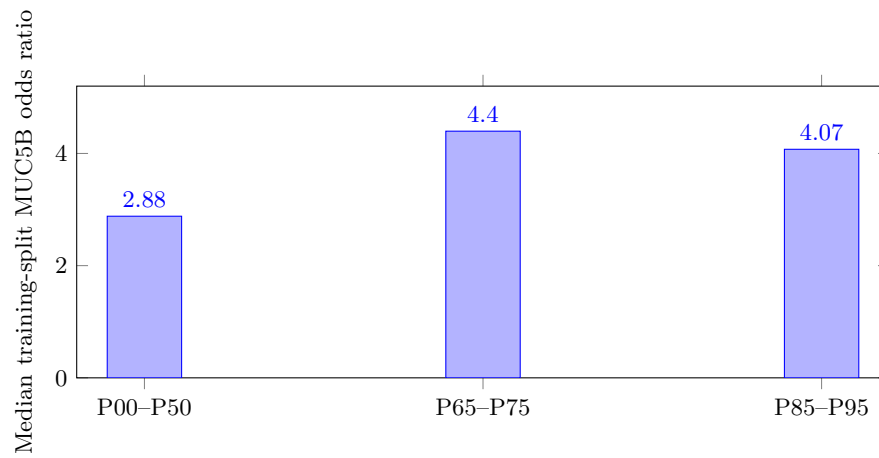


Fig. 3. Training-split MUC5B disease association in the three fixed ZeBRA-percentile switch regions used for the exploratory boundary rule. Median odds ratios across 100 splits were 2.88 (P00-P50), 4.40 (P65-P75), and 4.07 (P85-P95).